

NINGBO, China

WMO: 582390

Lat: 29.8153N

Lon: 121.4740E

Elev: 13

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	30.3	12.0	10.2	40.7	17.3	13.2	40.6	18.5	47.2	16.4	45.6	3.6	310	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	98.8	81.5	96.5	81.2	93.4	80.7	83.3	93.3	82.4	91.5	81.6	89.7	7.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.7	86.5	80.5	158.5	86.4	79.1	151.3	85.5	47.0	93.2	46.2	92.6	45.3	90.6	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.6	15.5	13.7	23.9	103.1	2.8	1.9	21.9	104.5	20.2	105.6	18.6	106.7	16.5	108.1		
			22.4	85.1	3.3	0.9	20.1	85.8	18.2	86.3	16.3	86.8	14.0	87.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	42.5	44.9	52.7	62.7	71.3	76.8	85.3	77.2	67.9	57.4	46.1		
		DBStd	15.99	6.50	6.77	7.32	6.74	5.20	5.13	4.75	3.97	4.77	4.86	6.43		
		HDD50	646	248	172	54	2	0	0	0	0	0	16	154		
		HDD65	2626	697	563	386	121	9	0	0	0	24	241	585		
		CDD50	5880	17	29	138	383	661	803	1111	1095	817	554	238		
		CDD65	2384	0	0	5	52	205	353	646	630	367	113	0		
		CDH74	23566	2	1	50	365	1316	2649	8068	7707	2877	480	51		
		CDH80	10174	0	0	3	75	387	931	4121	3702	874	74	7		
(14)	Wind	WSAvg	6.3	6.1	6.1	6.6	6.8	6.6	5.6	7.2	6.9	5.7	6.1	5.5	6.0	
(15)	Precipitation	PrecAvg	58.4	3.0	3.0	4.8	4.4	5.0	8.9	5.6	8.2	6.7	3.1	3.2	2.4	
		PrecMax	76.5	8.5	5.8	10.9	9.8	8.8	16.9	13.2	14.1	14.3	10.8	9.3	6.7	
		PrecMin	36.0	0.9	0.6	1.6	1.8	1.9	2.9	0.9	2.2	2.2	0.2	0.8	0.1	
		PrecStd	8.7	1.5	1.5	2.6	1.9	2.0	3.3	2.8	3.1	2.9	2.6	2.2	1.8	
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	70.9	78.9	86.3	91.8	96.9	102.6	102.2	93.4	87.5	80.2	68.4	
			MCWB	56.1	58.5	62.9	67.3	74.8	82.0	81.5	81.8	79.5	78.0	68.3	59.6	
		2%	DB	62.4	65.9	74.8	82.2	87.9	92.8	100.1	98.6	91.0	81.0	74.8	64.1	
			MCWB	53.4	56.6	62.3	66.6	71.9	81.0	81.7	81.2	78.2	70.7	66.3	56.6	
		5%	DB	57.1	60.9	69.7	78.6	84.3	89.2	97.1	95.4	87.8	78.7	70.2	59.3	
			MCWB	50.3	54.3	58.9	65.7	70.6	78.3	81.4	80.7	77.2	68.1	63.9	52.9	
		10%	DB	53.2	56.9	65.8	74.9	80.8	85.6	94.8	93.2	84.6	76.7	67.9	57.0	
			MCWB	48.0	51.3	57.5	64.5	69.0	77.2	81.1	80.9	75.7	67.3	62.3	51.6	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.8	62.6	68.1	73.5	79.0	83.5	84.5	84.3	81.5	78.9	72.8	64.3	
			MCDB	64.1	65.5	74.6	80.0	89.3	92.7	96.0	95.0	88.6	84.0	76.6	66.1	
		2%	WB	55.6	58.6	63.8	71.1	76.1	81.7	83.4	83.2	80.0	75.8	69.5	58.3	
			MCDB	59.7	63.4	70.2	76.6	82.6	88.9	94.1	92.8	85.9	78.0	71.7	61.0	
		5%	WB	52.2	55.5	61.2	69.0	74.5	80.1	82.5	82.4	79.1	72.5	66.6	55.6	
			MCDB	56.4	60.1	67.8	74.1	80.0	86.6	92.5	90.9	84.4	75.9	68.7	58.5	
		10%	WB	48.5	51.7	58.5	66.7	72.7	78.7	81.7	81.5	78.1	69.9	64.1	52.2	
			MCDB	51.5	55.8	63.9	72.1	77.2	83.5	90.8	89.2	83.0	73.8	66.7	54.9	
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.0	13.2	15.8	17.3	15.6	11.9	14.9	13.9	12.6	14.4	13.9	14.6	
			MCDBR	22.6	23.3	24.1	23.5	21.7	17.8	19.0	17.9	16.0	17.7	19.1	20.8	
		5% WB	MCWBR	14.3	14.2	13.1	11.1	9.2	7.0	5.9	5.4	6.0	8.0	10.1	12.4	
			MCWBR	18.6	18.9	20.2	17.3	15.4	14.9	17.2	15.3	13.5	11.6	12.3	14.3	
(40)	Clear-Sky Solar Irradiance	taub		0.614	0.656	0.699	0.708	0.675	0.693	0.645	0.658	0.632	0.610	0.605	0.590	
			taud	1.631	1.578	1.532	1.565	1.666	1.674	1.795	1.752	1.776	1.736	1.702	1.683	
		Ebn at Noon		188	196	200	206	214	209	219	214	213	205	189	185	
			Edh at Noon	69	79	88	88	79	78	69	72	68	66	63	62	
(44)	All-Sky Solar Radiation	RadAvg		704	819	1125	1340	1426	1280	1779	1556	1211	1075	771	714	
		RadStd		112	181	102	156	177	185	197	178	147	160	108	125	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+142	+92

Nomenclature: See separate page